

Human-Centered AI: Compliance & Responsible Innovation

2025-2026 Academic Year - Project Report

Group Information

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Project Title

Drowsiness Detection Using Convolutional Neural Networks (CNN)

1. Introduction

The project focuses on building a **deep learning** model to detect whether a person's eyes are open or closed from grayscale images using a Convolutional Neural Network (CNN).

Aim of the project:

- Build and train a CNN model that can accurately classify grayscale images into two categories.
- Improve generalization using **data augmentation**, **dropout**, and **early stopping**.

Importance / Problem addressed:

Fatigue and drowsiness (in driving/work environments) can lead to serious safety risks. An AI-based classifier can support early detection and help reduce accidents, provided it is designed responsibly (privacy, fairness, transparency).

2. Use of AI and Responsible Innovation Technologies

Which AI/IoT tools were used?

- Programming Language: Python.
- Frameworks: TensorFlow and Keras.
- Data Handling: split-folders library for ethical data splitting, and ImageDataGenerator for augmentation.
- Environment: Google Colab.

How were these technologies integrated responsibly?

We ensured responsible integration by prioritizing Data Integrity. Instead of blindly trusting high accuracy metrics, we audited our data pipeline. We used the split-folders library to physically separate video frames by subject identity before training. This ensures the model learns generalized features rather than memorizing specific faces, adhering to principles of robust and reliable AI.

What role did they play in solving the project's problem?

The CNN model acts as the core prediction engine. It automatically extracts features (edges, shapes of the eyelid) from grayscale images to determine the state of the eye without manual feature engineering.

How was ethical and responsible innovation embedded in the design? We prioritized Safety and Reliability. By fixing the data leakage issue, we ensured the system doesn't give a false sense of security (e.g., working perfectly on training data but failing on new drivers). Furthermore, the system is designed to process images locally (privacy-preserving) rather than storing user footage.

```
deep learning - project.ipynb
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# 4. Execute Split: 80% Training, 20% Validation
# Generates 'output_dataset/train' and 'output_dataset/val'
splitfolders.ratio(real_base_dir, output='output_dataset', seed=1337, ratio=(.8, .2), group_prefix=None)

# Define paths for downstream generators
train_dir = 'output_dataset/train'
val_dir = 'output_dataset/val'
print("\n👉 Success: Dataset split complete. Ready for model training.")

else:
    print("❌ Error: No valid image structure found.")
    print("Directory tree for debugging:")
    for root, dirs, files in os.walk('dataset_root'):
        level = root.replace('dataset_root', '').count(os.sep)
        print(f"{' ' * 4 * level}[os.path.basename(root)]/")
```

```
# --- Training Generator (With Augmentation) ---
# Apply data augmentation to improve generalization and prevent overfitting
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    width_shift_range=0.2,
    height_shift_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode='nearest'
)

train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=(64, 64), # Ensure this matches the model input shape
    batch_size=32,
    class_mode='binary', # Use 'categorical' if more than 2 classes
    color_mode='grayscale', # Set to grayscale based on model requirements
    shuffle=True
)
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

# --- Model Architecture Definition ---
model = Sequential([
    # Convolutional Block 1
    Conv2D(32, (3, 3), activation='relu', input_shape=(64, 64, 1)),
    MaxPooling2D(2, 2),
    Dropout(0.25), # Dropout to reduce overfitting

    # Convolutional Block 2
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),
    Dropout(0.25),

    # Convolutional Block 3
    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),
    Dropout(0.4), # Higher dropout for deeper layers

    # Fully Connected Layers
    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5), # 50% dropout in the dense layer
    Dense(1, activation='sigmoid') # Sigmoid activation for binary output (0 or 1)
])
```

3. Project Implementation and Development Process

What were the project steps?

1. **Data Collection:** Acquired a dataset of eye images (Open/Closed).
2. **Data Auditing (Critical Step):** Identified a "Data Leakage" issue where similar video frames appeared in both Train and Validation sets, causing false 100% accuracy.
3. **Smart Preprocessing:** Implemented split-folders to separate data by Video ID.
4. **Data Augmentation:** Applied Rotation, Zoom, and Shift transformations to the training set to improve robustness.
5. **Model Architecture:** Designed a Sequential CNN with 3 Convolutional blocks, Max Pooling, and Dropout layers.

6. **Training & Evaluation:** Trained the model using the Adam optimizer and evaluated it using a Confusion Matrix.

What sources, tools, and techniques were used?

- **Techniques:** Convolutional Neural Networks (CNN), Data Augmentation, Early Stopping (to prevent overfitting), Regularization using Dropout.
- **Tools:** Python libraries (NumPy, Matplotlib, Seaborn, Scikit-learn).

What challenges were encountered, and how were they resolved?

- **Challenge:** The initial model achieved 100% accuracy immediately, which was statistically improbable ("Too Good To Be True"). This was due to the model seeing identical frames in the validation set.
- **Resolution:** We restructured the dataset pipeline. Instead of a random shuffle split, we used grouped splitting based on the video source. We also introduced heavy data augmentation on the training set while keeping the validation set raw/clean.

How were the results evaluated? We evaluated the model using **Accuracy** and **Loss curves**, ensuring they converged naturally. The final validation was done using a **Confusion Matrix** on 800 unseen images, achieving a **99.87% accuracy** (only 1 error out of 800 samples), confirming the model's reliability

4. Ethical and Societal Dimensions

Was the AI/IoT system used ethically and responsibly?

Yes. The system is developed as an assistive safety tool, not a replacement for human responsibility. It respects the "Human-in-the-loop" principle by acting as a warning system.

What measures were taken regarding data privacy, bias, and transparency?

- **Privacy:** The design supports processing data on the edge (device-level) to minimize data transmission.
- **Bias Mitigation:** By using Data Augmentation (rotation, zoom), we reduced the bias related to head positioning or camera angle.
- **Transparency:** We documented the model architecture and the data splitting logic clearly to ensure the results are reproducible and not misleading.

Did the project consider sustainability and societal impact?

- **Societal Impact:** Directly contributes to road safety and public health.
- **Sustainability:** The model is optimized (small architecture) to run on low-power devices (like Raspberry Pi or mobile phones), reducing energy consumption compared to massive foundational models.

5. Compliance and Governance Roadmap

Ethical and Legal Principles Followed

Transparency:

Model limitations, data preprocessing steps, and evaluation procedures are clearly documented to ensure transparency and reproducibility.

Non-maleficence:

The system is designed to reduce the risk of harm by reducing the risk of false negatives and avoiding overconfident predictions. The model is intended as a decision-support tool rather than a fully autonomous decision-maker.

Privacy (GDPR Compliance):

Data minimization principles were followed by using grayscale images with low resolution and avoiding the storage of unnecessary personal or biometric information.

Generalization of the Roadmap for Similar Projects

Mandatory Data Audit:

AI systems based on video frames should include a mandatory data auditing step to detect and prevent frame-level data leakage between training and validation sets.

Privacy-First Design:

Where possible, models should be deployed on-device to avoid cloud storage and transmission of sensitive biometric data, reducing privacy risks.

Fail-Safe Mechanisms:

The system should include fail-safe behaviors such as alerting the user when camera input is obstructed or when prediction confidence levels are low. This aligns with the EU AI Act requirements for high-risk AI systems in transportation-related applications.

6. References

Patil, P. V. (2020). *MRL Eye Dataset (Drowsiness Detection)*. Kaggle.
<https://www.kaggle.com/datasets/prasadvpatil/mrl-dataset>

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<https://www.tensorflow.org/tutorials/images/cnn>

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TensorFlow Developers. (n.d.). *EarlyStopping callback*.
https://www.tensorflow.org/api_docs/python/tf/keras/callbacks/EarlyStopping

Scikit-learn Developers. (n.d.). *Model evaluation metrics*.
https://scikit-learn.org/stable/modules/model_evaluation.html

European Union. (n.d.). *General Data Protection Regulation (GDPR)*.
<https://gdpr.eu/>

European Commission. (n.d.). *EU Artificial Intelligence Act*.
<https://digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence>

Evaluation Criteria for AI Technology Use (Report):

- **Excellent:** AI tools correctly and effectively used, with detailed and technical explanations. Tools are perfectly integrated into the project.
 - **Good:** AI tools are correctly used, but more details and applications could be added.
 - **Average:** Tools are adequately used, but explanations are incomplete or superficial.
 - **Weak:** AI tool uses insufficient or incorrect information, limited contribution to the project.
 - **Fail:** AI tools not used or used incorrectly.
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Plagiarism Policy:

- All projects will be checked using plagiarism detection software designated by the university (e.g., Turnitin, iThenticate).
- Content entirely generated by AI with no student contribution will be graded low.
- Reports and presentations must clearly indicate the extent of AI-generated output; the production process must be explained transparently.
- All project documents (report, presentation, video link) must be submitted before the final class of the semester.

Evaluation of Rubric for Human-Centered AI: Compliance & Responsible Innovation Projects (100 points)

Criterion	Description	Weight (%)	Excellent	Good	Average	Weak	Fail
Problem Definition & Relevance	Has the team defined a clear, feasible, and original ethical AI/Responsible Innovation problem aligned with the project themes (bias auditing, ethical IoT, compliance roadmap)?	15%	Clearly defined, original, and highly relevant ethical AI/Responsible Innovation problem with strong societal importance.	Defined problem relevant to ethical AI or responsible innovation but with limited originality or depth.	The problem is somewhat related but lacks depth or clarity.	The problem is vague, weakly connected to ethical AI or responsible innovation.	No problem defined or irrelevant to the course themes.
Integration of Responsible Innovation Frameworks	Were ethical principles, fairness, transparency, and accountability frameworks integrated effectively?	20%	Strong integration of ethical frameworks (e.g., fairness metrics, GDPR, EU AI Act, IEEE standards) with clear justification and application.	Relevant frameworks are considered, but integration lacks depth or full application.	Frameworks are mentioned but not applied meaningfully.	Minimal mention of frameworks with superficial application.	No frameworks considered; work ignores ethical or legal context.
Student Contribution & Critical Reflection	Beyond technical work, is the student's independent contribution and ethical	20%	Clear and critical reflection on both technical and ethical aspects,	Contribution and reflection are visible, though depth could be improved.	Some independent input, but mostly descriptive and shallow.	Minimal student contribution, weak reflection on ethics.	Work is entirely auto-generated or lacking personal contribution.
	reflection visible?		demonstrating strong independent thinking.		e and shallow.		contribution.

Bias & Impact Assessment	Did the project meaningfully assess bias, risks, or societal impact?	15%	Thorough analysis of bias, risks, and broader societal impact, with proposed mitigation strategies.	Bias/impact assessed with some depth, but limited mitigation.	Superficial mention of bias or impact; no mitigation.	Very weak or incomplete assessment of bias/impact.	No consideration of bias, risks, or impact.
Compliance & Governance Roadmap	Did the project propose actionable steps for compliance and governance (auditability, accountability, transparency)?	10%	Clear and actionable roadmap for compliance with strong alignment to real-world standards.	Roadmap proposed, but missing some clarity or depth.	Roadmap is partially relevant but incomplete.	Minimal or vague roadmap.	No compliance or governance considerations.
Source Citation & Transparency	Were all sources, frameworks, and AI/IoT tools clearly cited and documented?	10%	All tools, data, and frameworks cited clearly with proper formatting (APA, MLA, IEEE).	Most sources cited with minor errors.	Sources are incomplete or inconsistently cited.	Few sources cited, weak transparency.	No sources cited or transparency ignored.
Group Participation & Collaboration	Did group members contribute equally and collaborate effectively?	5%	Balanced and well-coordinated collaboration; clear evidence of equal participation.	Generally balanced participation with minor imbalances.	Some imbalance in contribution; collaboration is weak.	Very unbalanced participation; limited collaboration.	No evidence of collaboration.
Presentation & Communication	Was the final presentation/report engaging, clear, and professional?	5%	Highly professional, clear, engaging, with strong	Clear and structured, with some room for improvement in	Adequate but lacks clarity or strong delivery.	Weak presentation, poorly structured,	No presentation or report delivered.

			visuals and explanations.	engagement/visuals.		difficult to follow.	
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